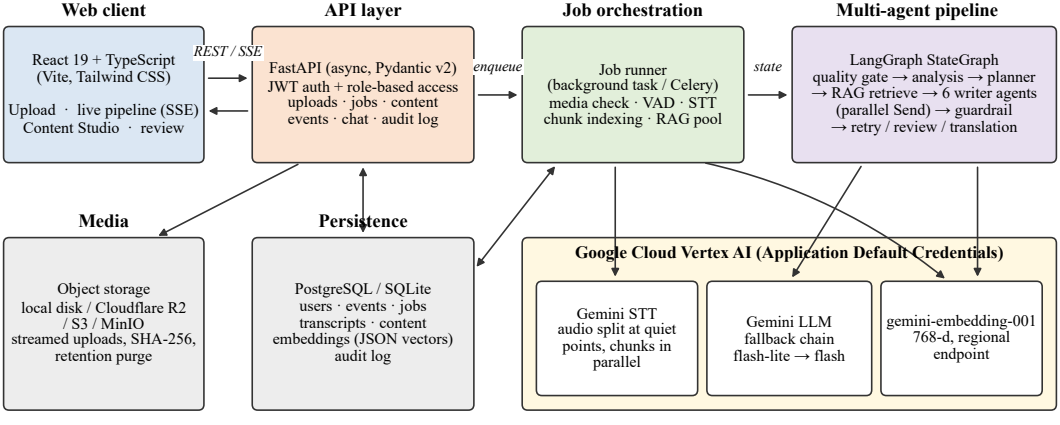

Web client
React 19 + TypeScript
(Vite, Tailwind CSS)
Upload · live pipeline (SSE)
Content Studio · review
REST / SSE
API layer
FastAPI (async, Pydantic v2)
JWT auth + role-based access
uploads · jobs · content
events · chat · audit log
enqueue
Job orchestration
Job runner
(background task / Celery)
media check · VAD · STT
chunk indexing · RAG pool
state
Multi-agent pipeline
LangGraph StateGraph
quality gate → analysis → planner
→ RAG retrieve → 6 writer agents
(parallel Send) → guardrail
→ retry / review / translation
Media
Object storage
local disk / Cloudflare R2
/ S3 / MinIO
streamed uploads, SHA-256,
retention purge
Persistence
PostgreSQL / SQLite
users · events · jobs
transcripts · content
embeddings (JSON vectors)
audit log
Google Cloud Vertex AI (Application Default Credentials)
Gemini STT
audio split at quiet
points, chunks in
parallel
Gemini LLM
fallback chain
flash-lite → flash
gemini-embedding-001
768-d, regional
endpoint